



KTU NOTES

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NOTIFICATIONS | SOLVED QUESTION PAPERS**

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JULY 2016

1. Form the p.d.e by eliminating the arbitrary function from
 $z = y^2 + 2f\left(\frac{1}{x} + \log y\right)$

Ans: Refer NB, Module 1, pg no: 18

2. Solve $p\sqrt{x} + q\sqrt{y} = \sqrt{z}$

Ans: This is of the form $Pp + Qq = R$ where $P = \sqrt{x}$, $Q = \sqrt{y}$, $R = \sqrt{z}$

$$A.C \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}}$$

$$\frac{dx}{\sqrt{x}} = \frac{dy}{\sqrt{y}} \Rightarrow \int \frac{dx}{\sqrt{x}} = \int \frac{dy}{\sqrt{y}} \Rightarrow 2\sqrt{x} = 2\sqrt{y} + c$$

$$\Rightarrow 2(\sqrt{x} - \sqrt{y}) = c \Rightarrow \sqrt{x} - \sqrt{y} = \frac{c}{2}$$

$$\Rightarrow \sqrt{x} - \sqrt{y} = a, \text{ where } u = \sqrt{x} - \sqrt{y}$$

$$\frac{dy}{\sqrt{y}} = \frac{dz}{\sqrt{z}} \Rightarrow \int \frac{dy}{\sqrt{y}} = \int \frac{dz}{\sqrt{z}} \Rightarrow 2\sqrt{y} = 2\sqrt{z} + c$$

$$\Rightarrow 2(\sqrt{y} - \sqrt{z}) = c \Rightarrow \sqrt{y} - \sqrt{z} = \frac{c}{2}$$

$$\Rightarrow \sqrt{y} - \sqrt{z} = b, \text{ where } v = \sqrt{y} - \sqrt{z}$$

$\therefore$ solution is, $f(u, v) = 0$, i.e., $f(\sqrt{x} - \sqrt{y}, \sqrt{y} - \sqrt{z}) = 0$

3. Using the method of separation of variables solve $\frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u$,
 where $u(x, 0) = 3e^{-5x}$

Ans: ~~Refer NB, Module 1, pg no:~~

Let $u = XT$

$$\frac{\partial u}{\partial x} = X'T \quad ; \quad \frac{\partial u}{\partial t} = XT'$$

$$\therefore \frac{\partial u}{\partial x} = 2\frac{\partial u}{\partial t} + u \Rightarrow X'T = 2XT' + XT$$

$$X'T = 2XT' + XT = (2T' + T)X$$

$$\frac{X'}{X} = \frac{2T' + T}{T} = k \text{ (say)}$$

Case 1:

$$\frac{X'}{X} = k \Rightarrow X' = kX$$

$$\text{Let } X' = m, \quad X = 1 \Rightarrow m = k$$

$$\therefore X = C_1 e^{mx} = C_1 e^{kx}$$

Case 2:

$$\frac{2T' + T}{T} = k \Rightarrow 2T' + T = kT$$

$$\text{Let } T' = m, \quad T = 1 \Rightarrow 2m + 1 = k$$

$$\Rightarrow m = \frac{k-1}{2}$$

$$\therefore T = C_2 e^{mt} = C_2 e^{\frac{1}{2}(k-1)t}$$

$$\therefore \text{ soln is, } U = XT = C_1 e^{kx} C_2 e^{\frac{1}{2}(k-1)t}$$

$$\text{i.e., } U(x,t) = C_1 C_2 e^{kx} e^{\frac{1}{2}(k-1)t}$$

$$\text{Given, } U(x,0) = 3e^{-5x}$$

$$\text{We've, } U(x,0) = C_1 C_2 e^{kx}$$

$$\therefore 3e^{-5x} = C_1 C_2 e^{kx}$$

$$\text{Comparing, we get, } C_1 C_2 = 3, \quad k = -5$$

$$\therefore U(x,t) = 3e^{-5x} e^{\frac{1}{2}(-5-1)t}$$

$$U(x,t) = 3e^{-5x} e^{-3t} = 3e^{-(5x+3t)}$$

4. State the one dimensional wave equation with boundary conditions & initial conditions for solving it

Ans:

$$\text{One dimensional wave equation is, } \frac{\partial^2 y}{\partial t^2} = c^2 \frac{\partial^2 y}{\partial x^2}$$

Initial Conditions

$$y(x,0) = f(x), \quad 0 < x < l$$

$$\left(\frac{\partial y}{\partial t} \right)_{t=0} = 0$$

Boundary conditions

$$y(0,t) = 0$$

$$y(l,t) = 0$$

where l is the length of the string

If the roots are real,
 $m_1, m_2, \dots, m_n$
(all distinct) then soln

$$\text{is, } C_1 e^{m_1 x} + C_2 e^{m_2 x} + \dots + C_n e^{m_n x}$$

5. In the heat equation, $\frac{\partial u}{\partial t} = \alpha^2 \frac{\partial^2 u}{\partial x^2}$, what does α^2 indicate? State the boundary & initial conditions for solving it.

Ans: α^2 indicates the diffusivity of the material of the bar.

Boundary condition

$$u(0, t) = u(l, t) = 0$$

Initial condition

$$u(x, 0) = f(x), \quad 0 < x < l$$

where l is the length of the bar.

6. Solve $(x+y)zp + (x-y)zq = x^2+y^2$

Ans: This is of the form $Pp + Qq = R$ where $P = (x+y)z$, $Q = (x-y)z$

$$\& R = x^2+y^2$$

$$\text{A-C} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{(x+y)z} = \frac{dy}{(x-y)z} = \frac{dz}{x^2+y^2}$$

$$\text{Let } P_1 = -x, \quad Q_1 = y, \quad R_1 = z$$

$$\begin{aligned} \text{Then } PP_1 + QQ_1 + RR_1 &= (x+y)z(-x) + (x-y)z(y) + (x^2+y^2)z \\ &= -x^2z - xyz + xyz - y^2z + x^2z + y^2z \\ &= 0 \end{aligned}$$

$$\text{Form, } P_1 dx + Q_1 dy + R_1 dz = 0$$

$$\Rightarrow -x dz + y dy + z dz = 0$$

$$\text{Integrating, } -\frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C_1 \Rightarrow -x^2 + y^2 + z^2 = 2C_1$$

$$\text{ie, } -x^2 + y^2 + z^2 = a, \quad \text{where } U = -x^2 + y^2 + z^2$$

$$\text{Let } P_2 = y, \quad Q_2 = \frac{x}{y}, \quad R_2 = -z$$

$$\begin{aligned} \text{Then } PP_2 + QQ_2 + RR_2 &= (x+y)z \frac{y}{y} + (x-y)z \frac{x}{y} - (x^2+y^2)z \\ &= xy z + \frac{x^2 z}{y} - \frac{xy^2 z}{y} - x^2 z - y^2 z = 0 \end{aligned}$$

$$\text{Form, } P_2 dx + Q_2 dy + R_2 dz = 0$$

$$\Rightarrow y dx + \frac{x}{y} dy - z dz = 0$$

$$\text{Integrating, } \frac{xy}{y} - \frac{z^2}{2} = C_2 \quad d(xy) - z dz = 0$$

$$\text{Integrating, } xy - \frac{z^2}{2} = b, \quad \text{where } V = xy - \frac{z^2}{2}$$

$$\therefore \text{ solution is } f(U, V) = 0, \quad \text{ie, } f(-x^2 + y^2 + z^2, xy - \frac{z^2}{2}) = 0$$

7. A lightly stretched string with fixed end points $x=0$ and $x=l$ is initially in a position given by $y = y_0 \sin^3\left(\frac{\pi x}{l}\right)$. If it is released from rest from this position, find the displacement $y(x, t)$.

Ans: Refer NB, pg no: 15

8. A lightly stretched string with fixed end points $x=0$ & $x=l$ is initially at rest in its equilibrium position. If it is vibrating by giving to each of its points a velocity $\lambda x(l-x)$, find the displacement of the string from one end at any time t .

Ans: Refer NB, pg no: 12

MODEL QUESTION PAPER

PDE

AUGUST 2016

1. Obtain the pde by eliminating the arbitrary functions from $x^2 + y^2 + z^2 = f(xy)$

Ans: $x^2 + y^2 + z^2 = f(xy)$

Diff. partially wrt x , $2x + 2z p = f'(xy) y$ — (1)

Diff. partially wrt y , $2y + 2z q = f'(xy) x$ — (2)

$$\frac{(1)}{(2)} \Rightarrow \frac{2x + 2z p}{2y + 2z q} = \frac{f'(xy) y}{f'(xy) x}$$

$$\frac{x + zp}{y + zq} = \frac{y}{x} \Rightarrow \underline{x(x + zp) = y(y + zq)}$$

2. Solve $y^2 z p + x^2 z q = xy^2$

Ans: This is of the form $Pp + Qq = R$
where $P = y^2 z$, $Q = x^2 z$, $R = xy^2$

$$\therefore \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\Rightarrow \frac{dx}{y^2 z} = \frac{dy}{x^2 z} = \frac{dz}{xy^2}$$

$$\frac{dx}{y^2 z} = \frac{dy}{x^2 z} \Rightarrow \frac{dx}{y^2} = \frac{dy}{x^2} \Rightarrow x^2 dx = y^2 dy$$

Integrating, $\frac{x^3}{3} = \frac{y^3}{3} + c$
 $\Rightarrow \frac{x^3}{3} - \frac{y^3}{3} = c \Rightarrow x^3 - y^3 = 3c \Rightarrow x^3 - y^3 = a$
 $\Rightarrow u = x^3 - y^3$

$$\frac{dx}{y^2 z} = \frac{dz}{xy^2} \Rightarrow \frac{dx}{z} = \frac{dz}{x} \Rightarrow x dx = z dz$$

Integrating, $\frac{x^2}{2} = \frac{z^2}{2} + c \Rightarrow \frac{x^2}{2} - \frac{z^2}{2} = c$

$$\Rightarrow x^2 - z^2 = 2c \Rightarrow x^2 - z^2 = b$$

$$\Rightarrow V = x^2 - z^2$$

$\therefore$ solution is, $f(u, v) = 0$
 $\Rightarrow f(x^3 - y^3, x^2 - z^2) = 0$

3. Solve $u_x + u_y = 0$ by method of separation of variables

Ans: $u_x + u_y = 0 \Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = 0$

Let the solution be, $u(x, y) = X Y$

Then, $\frac{\partial u}{\partial x} = X' Y$ & $\frac{\partial u}{\partial y} = X Y'$

$$u_x + u_y = 0 \Rightarrow X' Y + X Y' = 0 \Rightarrow X' Y = -X Y'$$

$$\Rightarrow \frac{X'}{X} = \frac{-Y'}{Y} = k, \text{ (say).}$$

Case 1: $\frac{X'}{X} = k \Rightarrow X' = k X$
 $\Rightarrow m = k \Rightarrow X = C_1 e^{kx}$

Case 2: $\frac{-Y'}{Y} = k \Rightarrow -Y' = k Y$
 $\Rightarrow -m = k$
 $\Rightarrow m = -k \Rightarrow Y = C_2 e^{-ky}$

$\therefore$ solution is, $u(x, y) = C_1 e^{kx} \cdot C_2 e^{-ky}$
 ie, $u(x, y) = C_1 C_2 e^{k(x-y)}$

4. A finite string of length L is fixed at both ends & is released from rest with a displacement $f(x)$. What are the initial & boundary conditions involved in this problem?

Ans:

Boundary conditions

$$y(0, t) = 0$$

$$y(L, t) = 0$$

Initial conditions

$$y(x, 0) = f(x), \quad 0 \leq x \leq L$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0$$

5. Write all the possible solutions of 1-D heat transfer equation.

Ans:

$$U(x, t) = (C_1 e^{px} + C_2 e^{-px}) C_3 e^{c^2 p^2 t}$$

$$U(x, t) = (C_1 \cos px + C_2 \sin px) C_3 e^{-c^2 p^2 t}$$

$$U(x, t) = (C_1 x + C_2) C_3$$

- 6(a) Solve $(y+zx)p - (x+yz)q = x^2 - y^2$

- (b) Find the differential equation of all spheres of fixed radius having their centres in xy -plane.

Ans: (a) This is of the form $Pp + Qq = R$, where $P = y+zx$, $Q = -(x+yz)$, $R = x^2 - y^2$

$$\text{A.C.} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R}$$

$$\text{i.e., } \frac{dx}{y+zx} = \frac{dy}{-(x+yz)} = \frac{dz}{x^2 - y^2}$$

$$\text{Let } P_1 = x, \quad Q_1 = y, \quad R_1 = -z$$

$$PP_1 + QQ_1 + RR_1 = (y+zx)x + -(x+yz)y + (x^2 - y^2)(-z) \\ = xy + x^2z - xy - y^2z - x^2z + y^2z = 0$$

$$\text{Form, } P_1 dx + Q_1 dy + R_1 dz = 0$$

$$\Rightarrow x dx + y dy - z dz = 0$$

$$\text{Integrating, } \frac{x^2}{2} + \frac{y^2}{2} - \frac{z^2}{2} = C_1 \Rightarrow x^2 + y^2 - z^2 = 2C_1$$

$$\Rightarrow x^2 + y^2 - z^2 = a$$

$$U = a$$

$$\text{where } U = x^2 + y^2 - z^2$$

$$\text{Let } P_2 = x, \quad Q_2 = -y, \quad R_2 = z$$

$$PP_2 + QQ_2 + RR_2 = (y+zx)y + \cancel{xy} - (x+yz)x + (x^2-y^2) \\ = y^2 + xyx - x^2 - xyx + x^2 - y^2 = 0$$

$$\text{Form } P_2 dx + Q_2 dy + R_2 dz = 0$$

$$\Rightarrow ydx + xdy + dz = 0$$

$$d(xy) + dz = 0$$

$$\text{Integrating, } xy + z = b, \text{ i.e.; } V = b \text{ where } V = xy + z.$$

$$\therefore \text{ solution is, } \underline{f(V, V) = 0}, \text{ i.e.; } \underline{f(x^2 + y^2 - z^2, xy + z) = 0}$$

(b) Refer NB, Module 1, page no: 30 (plane $z=0$ & xy plane are the same)

7. A tightly stretched homogeneous string of length 20 cm with its fixed ends executes transverse vibrations. Motion starts with zero initial velocity by displacing the entire string in the form $f(x) = k(x^2 - x^3)$. Find the deflection $u(x, t)$ at any time t .

Ans: Solution to the wave eq. is,

$$u(x, t) = \frac{1}{\sqrt{\rho A}} (C_1 \cos px + C_2 \sin px) (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (1)}$$

Boundary Conditions

$$u(0, t) = 0 \quad \text{--- (2)}$$

$$u(20, t) = 0 \quad \text{--- (3)}$$

Initial Conditions

$$\left(\frac{\partial u}{\partial t}\right)_{t=0} = 0 \quad \text{--- (4)}$$

$$u(x, 0) = f(x) = k(x^2 - x^3) \quad \text{--- (5)}$$

$$\text{From (2), } C_1 = 0$$

$$\text{(1)} \Rightarrow u = C_2 \sin px (C_3 \cos cpt + C_4 \sin cpt) \quad \text{--- (6)}$$

$$\text{From (3), } p = \frac{n\pi}{20}$$

$$\text{(6)} \Rightarrow u = C_2 \sin \frac{n\pi x}{20} \left(C_3 \cos \frac{n\pi ct}{20} + C_4 \sin \frac{n\pi ct}{20} \right) \quad \text{--- (7)}$$

From (4),

$$\frac{\partial u}{\partial t} = C_2 \sin \frac{n\pi x}{20} \left(-C_3 \sin \frac{n\pi ct}{20} \cdot \left(\frac{n\pi c}{20}\right) + C_4 \cos \frac{n\pi ct}{20} \cdot \left(\frac{n\pi c}{20}\right) \right)$$

$$\left(\frac{\partial u}{\partial t}\right)_{t=0} = 0 \Rightarrow C_2 \sin \frac{n\pi x}{20} \left(0 + C_4 \cdot \frac{n\pi c}{20} \right) = 0 \Rightarrow C_4 = 0$$

∴ solution becomes,

$$U(x,t) = c_2 c_3 \sum_{n=1}^{\infty} \frac{n\pi x}{20} \cos \frac{n\pi ct}{20} = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{20}\right) \cos\left(\frac{n\pi ct}{20}\right)$$

By Fourier Series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right) \text{ where } b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx$$

From (5), $U(x,0) = k(x^2 - x^3)$

$$\Rightarrow U(x,0) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{20}\right) \Rightarrow \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{20}\right) = k(x^2 - x^3)$$

This is of the form of a Fourier Series, where

$$f(x) = k(x^2 - x^3) \text{ \& } l = 20$$

$$\therefore b_n = \frac{2}{20} \int_0^{20} k(x^2 - x^3) \sin\left(\frac{n\pi x}{20}\right) dx = \frac{k}{10} \int_0^{20} (x^2 - x^3) \sin\left(\frac{n\pi x}{20}\right) dx$$

$U = x^2 - x^3$	$V = \sin\left(\frac{n\pi x}{20}\right)$
$U'_x = 2x - 3x^2$	$V_1 = -\cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi}$
$U''_x = 2 - 6x$	$V_2 = -\sin\left(\frac{n\pi x}{20}\right) \cdot \left(\frac{20}{n\pi}\right)^2$
$U'''_x = -6$	$V_3 = \cos\left(\frac{n\pi x}{20}\right) \left(\frac{20}{n\pi}\right)^3$
$U^{IV}_x = 0$	$V_4 = \sin\left(\frac{n\pi x}{20}\right) \left(\frac{20}{n\pi}\right)^4$

$$\therefore b_n = \left[UV_1 - U'_1 V_2 + U''_1 V_3 - U'''_1 V_4 \right]_0^{20}$$

$$= \left[(x^2 - x^3) \cdot \cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi} + (2x - 3x^2) \sin\left(\frac{n\pi x}{20}\right) \cdot \left(\frac{20}{n\pi}\right)^2 + \right. \\ \left. (2 - 6x) \cos\left(\frac{n\pi x}{20}\right) \cdot \left(\frac{20}{n\pi}\right)^3 - 6 \sin\left(\frac{n\pi x}{20}\right) \left(\frac{20}{n\pi}\right)^4 \right]_0^{20}$$

$$= \left[\frac{-(20^3 - 20^3) \cos(n\pi) \cdot 20}{n\pi} + (40 - 3 \cdot (20)^2) \sin(n\pi) \cdot \left(\frac{20}{n\pi}\right)^2 + \right. \\ \left. (2 - 6(20)) \cos(n\pi) \left(\frac{20}{n\pi}\right)^3 - 6 \sin(n\pi) \left(\frac{20}{n\pi}\right)^4 \right] - \\ \left[0 + 0 + 2 \cos(0) \left(\frac{20}{n\pi}\right)^3 - 6 \sin(0) \left(\frac{20}{n\pi}\right)^4 \right]$$

$$b_n = 7600 (-1)^n \cdot \frac{20}{n\pi} - 118 (-1)^n \left(\frac{20}{n\pi}\right)^3 - 2 \left(\frac{20}{n\pi}\right)^3$$

$$\therefore U(x,t) = \sum_{n=1}^{\infty} \left[7600 (-1)^n \frac{20}{n\pi} - 118 (-1)^n \left(\frac{20}{n\pi}\right)^3 - 2 \left(\frac{20}{n\pi}\right)^3 \right] \sin\left(\frac{n\pi x}{20}\right) \cos\left(\frac{n\pi ct}{20}\right)$$

8. A tightly stretched string of length a with fixed ends is initially in equilibrium position. Find the displacement $U(x, t)$ of the string if it is set vibrating by giving each of its points a velocity $V_0 \sin^3\left(\frac{\pi x}{a}\right)$

Ans: Refer NB, Module-2, page no: 15.

9. Find the temperature distribution in a rod of length $2m$ whose end points are maintained at temperature zero & initial temperature $f(x) = 100(2x - x^2)$

Ans: Solution of heat equation is,

$$U(x, t) = (C_1 \cosh px + C_2 \sinh px) e^{-c^2 p^2 t}$$

Boundary conditions, $u(0, t) = u(2, t) = 0$ — (1)

Initial condition, $u(x, 0) = f(x) = 100(2x - x^2)$ — (2)

From (1), $C_1 = 0$

∴ solution becomes, $U(x, t) = C_2 \sinh px e^{-c^2 p^2 t}$

From (1), $p = \frac{n\pi}{2}$, $n \in \mathbb{Z}$.

∴ solution becomes, $U(x, t) = C_2 \sin\left(\frac{n\pi x}{2}\right) e^{-c^2 \left(\frac{n\pi}{2}\right)^2 t}$

i.e., $U(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right) e^{-\frac{c^2 n^2 \pi^2 t}{4}}$

From (2), $U(x, 0) = 100(2x - x^2)$

⇒ $100(2x - x^2) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{2}\right)$

From Fourier Sine series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{l}\right), \text{ where } b_n = \frac{2}{l} \int_0^l f(x) \sin\left(\frac{n\pi x}{l}\right) dx.$$

here $f(x) = 100(2x - x^2)$
 $l = 2$

∴ $b_n = \frac{2}{2} \int_0^2 100(2x - x^2) \sin\left(\frac{n\pi x}{2}\right) dx = 100 \int_0^2 (2x - x^2) \sin\left(\frac{n\pi x}{2}\right) dx.$

$= 100 \left[UV_1 - U'V_2 + U''V_3 \right]_0^2$

$= 100 \left[\dots \right]$

$$U = 2x - x^2$$

$$U' = 2 - 2x$$

$$U'' = -2$$

$$U''' = 0$$

$$V = \sin\left(\frac{n\pi x}{a}\right)$$

$$V_1 = -\cos\left(\frac{n\pi x}{a}\right) \cdot \frac{a}{n\pi}$$

$$V_2 = -\sin\left(\frac{n\pi x}{a}\right) \cdot \left(\frac{a}{n\pi}\right)^2$$

$$V_3 = \cos\left(\frac{n\pi x}{a}\right) \cdot \left(\frac{a}{n\pi}\right)^3$$

$$\text{ie, } b_n = 100 \left[(2x - x^2) \cos\left(\frac{n\pi x}{a}\right) \cdot \frac{a}{n\pi} + (2 - 2x) \sin\left(\frac{n\pi x}{a}\right) \cdot \left(\frac{a}{n\pi}\right)^2 - 2 \cos\left(\frac{n\pi x}{a}\right) \cdot \left(\frac{a}{n\pi}\right)^3 \right]_0^a$$

$$\text{ie, } b_n = 100 \left[0 + 0 - 2 \cos(n\pi) \cdot \left(\frac{a}{n\pi}\right)^3 - \left(0 + 0 - 2 \cos 0 \cdot \left(\frac{a}{n\pi}\right)^3 \right) \right]$$

$$b_n = 100 \left[-2(-1)^n \left(\frac{a}{n\pi}\right)^3 + 2 \left(\frac{a}{n\pi}\right)^3 \right] = 100 \times 2 \times \left(\frac{a}{n\pi}\right)^3 [1 - (-1)^n]$$

$$b_n = \frac{1600}{n^3 \pi^3} [1 - (-1)^n]$$

$$\therefore u(x, t) = \sum_{n=1}^{\infty} \frac{1600}{n^3 \pi^3} [1 - (-1)^n] \sin\left(\frac{n\pi x}{a}\right) e^{-\frac{c^2 n^2 \pi^2 t}{4}}$$

MODEL QUESTION PAPER

PARTIAL DIFFERENTIALS

JULY 2018

1. Find pde representing the family of spheres whose centers lies on Z-axis.

Ans: Refer NB, Module -1, page no: 30.

2. Write any 3 assumptions involved in the derivation of 1-D wave equation.

Ans: Refer NB, Module -2, page no: 2 (write any 3)

3. Find the general solution of $(y^2 + z^2)p - xyq = -xz$

Ans: This is of the form $Pp + Qq = R$,

where $P = y^2 + z^2$, $Q = -xy$, $R = -xz$

$$A.C \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{y^2+z^2} = \frac{dy}{-xy} = \frac{dz}{-xz}$$

$$\frac{dy}{-xy} = \frac{dz}{-xz}$$

$$\Rightarrow \frac{dy}{y} = \frac{dz}{z}$$

$$\Rightarrow \log y = \log z + \log a$$

$$\Rightarrow \log y - \log z = \log a \Rightarrow \log \left(\frac{y}{z} \right) = \log a$$

$$\Rightarrow \frac{y}{z} = a, \text{ i.e., } U = a \text{ where } U = \frac{y}{z}$$

$$\text{Let } P_1 = x, Q_1 = y, R_1 = z$$

$$\text{Then } PP_1 + QQ_1 + RR_1 = (y^2+z^2)x + (-xy)y + (-xz)z \\ = xy^2 + xz^2 - xy^2 - xz^2 = 0.$$

$$\text{Consider } P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow x dx + y dy + z dz = 0$$

$$\text{Integrating, } \frac{x^2}{2} + \frac{y^2}{2} + \frac{z^2}{2} = C, \Rightarrow x^2 + y^2 + z^2 = 2C,$$

$$\Rightarrow x^2 + y^2 + z^2 = b$$

$$\text{i.e., } V = b, \text{ where } V = x^2 + y^2 + z^2$$

$$\therefore \text{ general solution is, } f(U, V) = 0$$

$$\text{i.e., } f\left(\frac{y}{z}, x^2 + y^2 + z^2\right) = 0$$

4. Using method of separation of variables, Solve $\frac{\partial u}{\partial x} = 2 \frac{\partial u}{\partial t} - u$,
 $u(x, 0) = 5e^{-3x}$

Ans: Refer Qno: 3, July 2016.

5. A tightly stretched string of length l fastened at both ends is initially in a position given by $y = kx$, $0 \leq x < l$. If it is released from rest from this position, find the displacement $y(x, t)$.

Ans: Solution is, $y(x, t) = (C_1 \cos px + C_2 \sin px)(C_3 \cos pt + C_4 \sin pt)$

Boundary conditions: $y(0, t) = y(l, t) = 0$ — ①

Initial conditions: $\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0$ — ②

$y(x, 0) = kx$ — ③

From ①, $c_1 = 0$.

solution becomes, $y(x,t) = c_2 \sin \frac{n\pi x}{l} (c_3 \cos \frac{n\pi c t}{l} + c_4 \sin \frac{n\pi c t}{l})$

From ①, $p = \frac{n\pi}{l}$

Soln becomes, $y(x,t) = c_2 \sin \frac{n\pi x}{l} (c_3 \cos \frac{n\pi c t}{l} + c_4 \sin \frac{n\pi c t}{l})$

From ②, $\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0$

$$\frac{\partial y}{\partial t} = c_2 \sin \frac{n\pi x}{l} \left[-c_3 \sin \left(\frac{n\pi c t}{l} \right) \cdot \frac{n\pi c}{l} + c_4 \cos \left(\frac{n\pi c t}{l} \right) \cdot \frac{n\pi c}{l} \right] = 0$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = 0 \Rightarrow c_2 \sin \left(\frac{n\pi x}{l} \right) c_4 \frac{n\pi c}{l} = 0$$

$$\Rightarrow c_4 = 0$$

Soln becomes,

$$y(x,t) = c_2 c_3 \sin \frac{n\pi x}{l} \cos \frac{n\pi c t}{l}$$

$$y(x,t) = \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} \cos \frac{n\pi c t}{l}$$

From ④, $y(x,0) = kx$

$$\Rightarrow \sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} = kx$$

By Fourier Sine series,

$$\sum_{n=1}^{\infty} b_n \sin \frac{n\pi x}{l} = f(x) \quad \text{where } b_n = \frac{2}{l} \int_0^l f(x) \sin \frac{n\pi x}{l} dx$$

here $f(x) = kx$

$$\therefore b_n = \frac{2}{l} \int_0^l kx \sin \frac{n\pi x}{l} dx = \frac{2k}{l} \int_0^l x \sin \frac{n\pi x}{l} dx = \frac{2k}{l} [uv_1 - u'v_2]_0^l$$

$$\begin{array}{lcl} u = x & \swarrow & v = \sin \frac{n\pi x}{l} \\ u' = 1 & \searrow & v_1 = -\cos \frac{n\pi x}{l} \\ u'' = 0 & \searrow & v_2 = -\sin \frac{n\pi x}{l} \end{array}$$

$$b_n = \frac{2k}{l} \left[-x \cos \frac{n\pi x}{l} + \sin \frac{n\pi x}{l} \right]_0^l = \frac{2k}{l} [(-l \cos n\pi + 0) - (0 + 0)]$$

$$b_n = \frac{2k}{l} \times -l(-1)^n = -(-1)^n 2k.$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} -(-1)^n 2k \sin \frac{n\pi x}{l} \cos \frac{n\pi ct}{l}$$

5.(a) Solve $xp - yq = y^2 - x^2$

(b) $p - 2q = 3x^2 \sin(y + 2x)$

Ans: (a) This is of the form $Pp + Qq = R$, where $P = x$, $Q = -y$, $R = y^2 - x^2$

$$\therefore \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{x} = \frac{dy}{-y} = \frac{dz}{y^2 - x^2}$$

$$\frac{dx}{x} = \frac{dy}{-y} \Rightarrow \log x = -\log y + \log a$$

$$\Rightarrow \log x + \log y = \log a$$

$$\Rightarrow \log xy = \log a \Rightarrow xy = a$$

$$\text{ie, } U = a, \text{ where } U = xy$$

Choose $P_1 = x$, $Q_1 = y$, $R_1 = 1$

$$PP_1 + QQ_1 + RR_1 = x(x) + y(y) + (y^2 - x^2)1 = x^2 - y^2 + y^2 - x^2 = 0.$$

Form, $P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow x dx + y dy + dz = 0$

$$\text{Integrating, } \frac{x^2}{2} + \frac{y^2}{2} + z = C_1 \Rightarrow x^2 + y^2 + 2z = 2C_1$$

$$\Rightarrow x^2 + y^2 + 2z = b$$

$$\text{ie, } V = b, \text{ where } V = x^2 + y^2 + 2z$$

$$\therefore \text{ solution is, } f(U, V) = 0$$

$$\text{ie, } \underline{\underline{f(xy, x^2 + y^2 + 2z) = 0}}$$

(b) Refer NB, Module 1, page no: 46.

6. Derive 1-D wave equation

Ans: Refer NB, Module 2, page no: 2

Model Questions

JULY 2017

PDE

1. Obtain the pde of $z = f(x^2 + y^2)$

Ans: Diff. partially wst x , $p = f'(x^2 + y^2) \cdot 2x$ — (1)

Diff. partially wst y , $q = f'(x^2 + y^2) \cdot 2y$ — (2)

$$\frac{(1)}{(2)} \Rightarrow \frac{p}{q} = \frac{f'(x^2 + y^2) \cdot 2x}{f'(x^2 + y^2) \cdot 2y} = \frac{x}{y}$$

$$\Rightarrow \underline{py - qx = 0}$$

2. Solve $xp + yq = 3z$

Ans: This is of the form $Pp + Qq = R$, where $P = x$, $Q = y$, $R = 3z$

$$A-C \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{x} = \frac{dy}{y} = \frac{dz}{3z}$$

$$\frac{dx}{x} = \frac{dy}{y} \Rightarrow \log x = \log y + \log a \Rightarrow \log x - \log y = \log a$$

$$\Rightarrow \log \left(\frac{x}{y} \right) = \log a \Rightarrow \frac{x}{y} = a, \text{ ie, } U = a$$

where $U = \frac{x}{y}$

$$\frac{dy}{y} = \frac{dz}{3z} \Rightarrow \log y = \frac{1}{3} \log z + \log b \Rightarrow 3 \log y - \log z = \log b$$

$$\Rightarrow \log y^3 - \log z = \log b$$

$$\Rightarrow \log \left(\frac{y^3}{z} \right) = \log b \Rightarrow \frac{y^3}{z} = b, \text{ ie, } V = b$$

where $V = \frac{y^3}{z}$

$$\therefore \text{ solution is, } f(U, V) = 0, \text{ ie, } \underline{f\left(\frac{x}{y}, \frac{y^3}{z}\right) = 0}$$

3. Using method of separation of variables solve $U_{xy} - U = 0$.

Ans: Let $U(x, y) = x^4$

$$U_x = x^4; \quad U_{xy} = x^4 y'$$

$$\text{ie, } x^4 y' - x^4 = 0$$

$$\cancel{x^4} y' = \cancel{x^4}$$

$$y' = 1$$

$$x^4 y' = x^4$$

$$\frac{y'}{y} = \frac{1}{x} = k$$

Case 1: $\frac{x'}{x} = k \Rightarrow x' = kx \Rightarrow m = k \Rightarrow x = C_1 e^{kx}$

Case 2: $\frac{y}{y'} = k \Rightarrow y = ky' \Rightarrow y' = \frac{1}{k} y \Rightarrow m = \frac{1}{k} \Rightarrow y = C_2 e^{\frac{1}{k} y}$

$\therefore$ soln is, $U = xy = C_1 C_2 e^{kx} \cdot e^{\frac{y}{k}}$

4. State 1-D heat equation with boundary conditions & initial conditions for solving it.

Ans: $\frac{\partial u}{\partial t} = c^2 \frac{\partial^2 u}{\partial x^2}$

Boundary conditions: $u(0, t) = u(l, t) = 0$ where l is the length of the bar.
Initial conditions: $u(x, 0) = f(x)$

5. Solve: $(mz - ny)p + (nx - lz)q = ly - mx$.

Ans: Refer NB, Module-1, page no: 54

6. Solve 1-D wave equation $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$ with boundary conditions $u(0, t) = u(l, t) = 0$ for all t & initial conditions $u(x, 0) = f(x)$, $\frac{\partial u}{\partial t} \Big|_{t=0} = g(x)$

Ans: Refer NB, Module-2, page no: 2

7. Derive 1-D heat equation

Ans: Refer NB, Module-2, page no: 34

8. A string of length 20cm fixed at both ends is displaced from its position of equilibrium, by each of its points an initial velocity given by $= \begin{cases} x, & 0 < x \leq 10 \\ 20-x, & 10 \leq x \leq 20 \end{cases}$, x being the distance from one end.

Determine the displacement at any subsequent time

Ans:

$$(C_1 \cos px + C_2 \sin px)(C_3 \cos p t + C_4 \sin p t) = y(x, t)$$

Boundary conditions: $y(0, t) = y(20, t) = 0$ — (1)

Initial conditions: $\left(\frac{\partial y}{\partial t}\right)_{t=0} = \begin{cases} x, & 0 < x \leq 10 \\ 20-x, & 10 \leq x \leq 20 \end{cases}$ — (2)

$$u(x, 0) = 0 \text{ — (3)}$$

From (1), $C_1 = 0$.

Soln becomes, $y(x, t) = C_2 \sin px (C_3 \cos p t + C_4 \sin p t)$

From (1), $p = \frac{n\pi}{20}$

Soln becomes, $y(x, t) = C_2 \sin \frac{n\pi x}{20} \left(C_3 \cos \frac{n\pi t}{20} + C_4 \sin \frac{n\pi t}{20} \right)$

From (3), $u(x, 0) = 0$

$$\Rightarrow C_2 \sin \frac{n\pi x}{20} C_3 = 0 \Rightarrow C_3 = 0$$

Soln becomes, $y(x, t) = C_2 C_4 \sin \frac{n\pi x}{20} \sin \frac{n\pi t}{20}$

$$y(x, t) = \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{20} \right) \sin \left(\frac{n\pi t}{20} \right)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{20} \right) \cos \left(\frac{n\pi t}{20} \right) \cdot \frac{n\pi}{20}$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} \Rightarrow \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{20} \right) \cdot \frac{n\pi}{20}$$

$$\text{ie, } \sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{20} \right) \cdot \frac{n\pi}{20} = \begin{cases} x, & 0 < x \leq 10 \\ 20-x, & 10 \leq x \leq 20 \end{cases}$$

$$\sum_{n=1}^{\infty} n b_n \sin \left(\frac{n\pi x}{20} \right) = \frac{20}{\pi} \begin{cases} x, & 0 < x \leq 10 \\ 20-x, & 10 \leq x \leq 20 \end{cases}$$

By Fourier Series series,

$$\sum_{n=1}^{\infty} b_n \sin \left(\frac{n\pi x}{20} \right), \quad b_n = \frac{2}{l} \int_0^l f(x) \sin \left(\frac{n\pi x}{l} \right) dx$$

here $b_n \rightarrow n b_n$

$$f(x) = \frac{20}{\pi} \begin{cases} x, & 0 < x \leq 10 \\ 20-x, & 10 \leq x \leq 20 \end{cases}$$

$$l = 20$$

$$n b_n = \frac{2}{20} \int_0^{20} f(x) \sin\left(\frac{n\pi x}{20}\right) dx = \frac{1}{10} \times \frac{20}{\pi c} \left[\int_0^{10} x \sin\left(\frac{n\pi x}{20}\right) dx + \int_{10}^{20} (20-x) \sin\left(\frac{n\pi x}{20}\right) dx \right]$$

$$b_n = \frac{2}{n\pi c} \left[\int_0^{10} x \sin\left(\frac{n\pi x}{20}\right) dx + \int_{10}^{20} (20-x) \sin\left(\frac{n\pi x}{20}\right) dx \right]$$

$u = x$
 $u' = 1$
 $u'' = 0$

$v = \sin\left(\frac{n\pi x}{20}\right)$
 $v_1 = -\cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi}$
 $v_2 = -\sin\left(\frac{n\pi x}{20}\right) \cdot \left(\frac{20}{n\pi}\right)^2$

$u = 20-x$
 $u' = -1$
 $u'' = 0$

$v = \sin\left(\frac{n\pi x}{20}\right)$
 $v_1 = -\cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi}$
 $v_2 = -\sin\left(\frac{n\pi x}{20}\right) \cdot \left(\frac{20}{n\pi}\right)^2$

$$b_n = \frac{2}{n\pi c} \left[\left(-x \cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi} + \sin\left(\frac{n\pi x}{20}\right) \left(\frac{20}{n\pi}\right)^2 \right) \Big|_0^{10} + \left(-(20-x) \cos\left(\frac{n\pi x}{20}\right) \cdot \frac{20}{n\pi} - \sin\left(\frac{n\pi x}{20}\right) \left(\frac{20}{n\pi}\right)^2 \right) \Big|_{10}^{20} \right]$$

$$= \frac{2}{n\pi c} \left[\left(-10 \cos\left(\frac{n\pi}{2}\right) \cdot \frac{20}{n\pi} + \sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2 - 0 \right) + \left(0 - (-10 \cos\left(\frac{n\pi}{2}\right) \cdot \frac{20}{n\pi} - \sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2 \right) \right]$$

$$b_n = \frac{2}{n\pi c} \left[\sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2 + \sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2 \right]$$

$$b_n = \frac{2}{n\pi c} \cdot 2 \sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2 = \frac{4 \sin\left(\frac{n\pi}{2}\right) \left(\frac{20}{n\pi}\right)^2}{n\pi c} = \frac{1600 \sin\left(\frac{n\pi}{2}\right)}{n^3 \pi^3 c}$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \frac{1600 \sin\left(\frac{n\pi}{2}\right)}{n^3 \pi^3 c} \sin\left(\frac{n\pi x}{20}\right) \sin\left(\frac{n\pi ct}{20}\right)$$

9. Derive 1-D heat equation

Ans: Refer WB, Module 2, page no: 34

10. Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature 0°C assuming that the initial temperature is $f(x) = \begin{cases} x, & 0 < x < L/2 \\ L-x, & L/2 < x < L \end{cases}$

Ans: Solution is, $u(x,t) = (c_1 \cos px + c_2 \sin px) e^{-c^2 p^2 t}$ — (1)
 $u(0,t) = u(L,t) = 0$ — (2)

$$u(x,0) = \begin{cases} x, & 0 \leq x < L/2 \\ L-x, & L/2 \leq x \leq L \end{cases} \quad \text{--- (3)}$$

From (2), $u(0,t) = 0$

* (1) $\Rightarrow c_1 = 0$

Solution becomes, $u(x,t) = c_2 \sin px e^{-c^2 p^2 t}$

From (3), $u(L,t) = 0$.

Then, $p = \frac{n\pi}{L}, n \in \mathbb{N}$

∴ soln becomes,

$$u(x,t) = c_2 \sin\left(\frac{n\pi x}{L}\right) e^{-c^2 \left(\frac{n\pi}{L}\right)^2 t}$$

$$\Rightarrow u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) e^{-c^2 \left(\frac{n\pi}{L}\right)^2 t}$$

From (3),

$$u(x,0) = \begin{cases} x, & 0 \leq x \leq L/2 \\ L-x, & L/2 \leq x \leq L \end{cases}$$

$$\text{i.e., } \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) = \begin{cases} x, & 0 \leq x \leq L/2 \\ L-x, & L/2 \leq x \leq L \end{cases}$$

By Fourier Sine series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right), \text{ where } b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx.$$

here $f(x) = \begin{cases} x, & 0 \leq x \leq L/2 \\ L-x, & L/2 \leq x \leq L \end{cases}$

$$b_n = \frac{2}{L} \left[\int_0^{L/2} x \sin\left(\frac{n\pi x}{L}\right) dx + \int_{L/2}^L (L-x) \sin\left(\frac{n\pi x}{L}\right) dx \right]$$

$u = x$	$v = \sin\left(\frac{n\pi x}{L}\right)$	$u = L-x$	$v = \sin\left(\frac{n\pi x}{L}\right)$
$u' = 1$	$v_1 = -\cos\left(\frac{n\pi x}{L}\right) \cdot \left(\frac{L}{n\pi}\right)$	$u' = -1$	$v_1 = -\cos\left(\frac{n\pi x}{L}\right) \cdot \left(\frac{L}{n\pi}\right)$
$u'' = 0$	$v_2 = -\sin\left(\frac{n\pi x}{L}\right) \cdot \left(\frac{L}{n\pi}\right)^2$	$u'' = 0$	$v_2 = -\sin\left(\frac{n\pi x}{L}\right) \cdot \left(\frac{L}{n\pi}\right)^2$

$$\therefore b_n = \frac{2}{L} \left[\left(-x \cos\left(\frac{n\pi x}{L}\right) \left(\frac{L}{n\pi}\right) + \sin\left(\frac{n\pi x}{L}\right) \left(\frac{L}{n\pi}\right)^2 \right) \Big|_0^{L/2} + \left(-(L-x) \cos\left(\frac{n\pi x}{L}\right) \left(\frac{L}{n\pi}\right) - \sin\left(\frac{n\pi x}{L}\right) \left(\frac{L}{n\pi}\right)^2 \right) \Big|_{L/2}^L \right]$$

$$b_n = \frac{2}{L} \left[\left(-\frac{1}{2} \cos\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right) + \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2 - 0 \right) + \left(0 - \left(-\frac{1}{2} \cos\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right) - \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2 \right) \right] \right]$$

$$b_n = \frac{2}{L} \left[\sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2 + \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2 \right]$$

$$b_n = \frac{2}{L} \cdot 2 \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2$$

$$b_n = \frac{4}{L} \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} \frac{4}{L} \sin\left(\frac{n\pi}{2}\right) \left(\frac{L}{n\pi}\right)^2 \sin\left(\frac{n\pi x}{L}\right) e^{-\frac{c^2 n^2 \pi^2 t}{L}}$$

Model Questions

Exam May 2017

PDE

1. Form the p.d.e from $z = x g(y) + y f(x)$

Ans: Refer NB, Module 1, page no: 20.

2. Solve $(y-z)p + (x-y)q = (z-x)$

Ans: This is of the form $Pp + Qq = R$ where $P = (y-z)$, $Q = (x-y)$, $R = (z-x)$

$$\text{A.C.} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{y-z} = \frac{dy}{x-y} = \frac{dz}{z-x}$$

$$\text{Choose } P_1 = 1, Q_1 = 1, R_1 = 1$$

$$P_1 P_1 + Q_1 Q_1 + R_1 R_1 = y-z + x-y + z-x = 0$$

$$\text{Form, } P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow dx + dy + dz = 0$$

$$\text{Integrating, } x+y+z = a, \text{ here } U = x+y+z$$

$$\text{Choose } P_2 = x, Q_2 = z, R_2 = y$$

$$P_2 P_1 + Q_2 Q_1 + R_2 R_1 = x(y-z) + z(x-y) + y(z-x) = xy - xz + zx - zy + yz - xy = 0$$

$$\text{Form, } P_2 dx + Q_2 dy + R_2 dz = 0 \Rightarrow x dx + z dy + y dz = 0$$

$$\Rightarrow x dx + d(zy) = 0$$

Integrating,

$$\frac{x^2}{2} + zy = b, \text{ here } V = \frac{x^2}{2} + zy$$

$$\therefore \text{ solution is } f(U, V) = 0$$

$$\text{i.e., } f(x+y+z, \frac{x^2}{2} + zy) = 0$$

3. Write down the important assumptions to derive 1-D wave equation

Ans: Refer NB, Module 2, page no. 2

4. Solve 1-D heat equation when $k > 0$

Ans: Refer NB, Module 2, page no. 37, Case 1

5. Write down the possible solutions of 1-D heat equation

Ans: Refer NB, Module 2, page no. 36, 37 (Write the 3 solutions obtained when $k > 0$, $k < 0$ & $k = 0$)

6. Solve $3U_x + 2U_y = 0$ with $U(x,0) = 4e^{-x}$ by method of separation of variables

Ans:

$$U = XY$$

$$U_x = X'Y \quad \& \quad U_y = XY'$$

$$3X'Y + 2XY' = 0 \quad \Rightarrow \quad 3X'Y = -2XY' \quad \Rightarrow \quad \frac{3X'}{X} = -\frac{2Y'}{Y} = -k$$

Case 1: $\frac{3X'}{X} = k \quad \Rightarrow \quad 3X' = kX$

put $X=1, X'=m \quad \Rightarrow \quad 3m = k \quad \Rightarrow \quad m = k/3$

$$\therefore X = C_1 e^{k/3 x}$$

Case 2: $-\frac{2Y'}{Y} = k \quad \Rightarrow \quad -2Y' = kY$

put $Y=1, Y'=m \quad \Rightarrow \quad -2m = k \quad \Rightarrow \quad m = -k/2$

$$\therefore Y = C_2 e^{-k/2 y}$$

$$\therefore U(x,y) = C_1 C_2 e^{k/3 x - k/2 y}$$

given, $U(x,0) = 4e^{-x} \quad \Rightarrow \quad C_1 C_2 e^{k/3 x} = 4e^{-x}$

$$\Rightarrow C_1 C_2 = 4$$

$$k/3 = -1 \quad \Rightarrow \quad k = -3$$

$$\therefore \text{soln is, } U(x,y) = 4e^{-x/3} e^{-3/2 y}$$

$$U(x,y) = 4e^{-x - 3/2 y}$$

7(a) Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$

(b) Solve $(x+y)zp + (x-y)zq = x^2 + y^2$

(c) Form the pde by eliminating a, b, c from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Ans: (a) Refer Module 1, Page no. 53.

(b) This is of the form $Pp + Qq = R$, where $P = (x+y)z$, $Q = (x-y)z$, $R = x^2 + y^2$

$$\text{A.C.} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{(x+y)z} = \frac{dy}{(x-y)z} = \frac{dz}{x^2 + y^2}$$

Choose $P_1 = x$, $Q_1 = -y$, $R_1 = -z$

$$\begin{aligned} PP_1 + QQ_1 + RR_1 &= (x+y)xz - (x-y)zy - (x^2 + y^2)z \\ &= x^2z + xyz - xyz + y^2z - x^2z - y^2z = 0 \end{aligned}$$

Consider $P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow xdx - ydy - zdz = 0$

Integrating, $\frac{x^2}{2} - \frac{y^2}{2} - \frac{z^2}{2} = c_1 \Rightarrow x^2 - y^2 - z^2 = 2c_1 = a$
 $\Rightarrow V = x^2 - y^2 - z^2$

Choose $P_2 = x+y$, $Q_2 = x-y$, $R_2 = z$

Consider $\frac{dx}{(x+y)z} = \frac{dy}{(x-y)z}$

$$\Rightarrow \frac{dx}{x+y} = \frac{dy}{x-y}$$

$$\Rightarrow (x-y)dx = (x+y)dy \Rightarrow xdx - ydy = xdy + ydx$$

$$\Rightarrow xdx - ydy - xdy - ydx = 0$$

$$\Rightarrow xdx - ydy - d(xy) = 0$$

Integrating, $\frac{x^2}{2} - \frac{y^2}{2} - xy = b \Rightarrow V = \frac{x^2}{2} - \frac{y^2}{2} - xy$

Solution is, $f(V, V) = 0 \Rightarrow f(x^2 - y^2 - z^2, \frac{x^2}{2} - \frac{y^2}{2} - xy) = 0$

(c) Refer NB, Module -1, page no: 14.

8. Solve pde $\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$. Boundary conditions are $u(0, t) = u(l, t) = 0, t \geq 0$

Initial conditions are $y(x, 0) = a \sin\left(\frac{\pi x}{l}\right)$ & $\frac{\partial y}{\partial t} = 0$ at $t = 0$.

Ans: Refer Module -2, page no: 8

MODEL QUESTIONS

PDE

APRIL 2018

1. Form a pde by eliminating the arbitrary function in,
 $xyz = \phi(x+y+z)$

Ans: Diff. partially w.r.t x , $y \frac{\partial}{\partial x} [xyz] = \phi'(x+y+z) [1+p]$ — (1)
 Diff. partially w.r.t y , $x \frac{\partial}{\partial y} [xyz] = \phi'(x+y+z) [1+q]$ — (2)
 $\frac{(1)}{(2)} \Rightarrow \frac{y \frac{\partial}{\partial x} [xyz]}{x \frac{\partial}{\partial y} [xyz]} = \frac{\phi'(x+y+z) [1+p]}{\phi'(x+y+z) [1+q]} \Rightarrow \frac{y \frac{\partial}{\partial x} [xyz]}{x \frac{\partial}{\partial y} [xyz]} = \frac{1+p}{1+q}$
 $\Rightarrow \underline{yp(1+q) = xq(1+p)}$ $y[xp+z](1+q) = x[yq+z](1+p)$

2. Solve 1-D wave equation for $k < 0$

Ans: Refer NB, module -2, page no: 5,6 (Case 2 & final answer)

3. Solve $\frac{\partial u}{\partial x} - 2 \frac{\partial u}{\partial y} - u = 0$, $u(x,0) = 6e^{-3x}$ using method of separation of variables.

Ans: Refer NB, module -1, page no: 74

4. Write down the possible solutions of 1-D heat equation

Ans: Refer NB, module -2, page 36 (Write only the solution in each case)

MODEL QUESTIONS

PDE

DECEMBER 2018

1. Find the pde of all spheres of fixed radius having their centres in the $x-y$ plane.

Ans: Refer NB, Module -1, page no: 30

2. Write any 3 assumptions involved in the derivation of 1-D wave equation

Ans: Refer NB, Module -2, page no: 2

3. ~~Solve~~ Write down the possible solutions of 1-D heat equation.

Ans: Refer NB, Module - 2, page no. 36 (Write the 3 solutions)

4. Solve $p - 2q = 3x^2 \sin(y + 2x)$

Ans: Refer NB, Module - 1, page no. 46

5. Solve $x(y-z)p + y(z-x)q = z(x-y)$

Ans: This is of the form $Pp + Qq = R$, where $P = x(y-z)$, $Q = y(z-x)$, $R = z(x-y)$

$$\text{A.C.} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{x(y-z)} = \frac{dy}{y(z-x)} = \frac{dz}{z(x-y)}$$

Choose $P_1 = \frac{1}{x}$, $Q_1 = \frac{1}{y}$, $R_1 = \frac{1}{z}$

$$PP_1 + QQ_1 + RR_1 = x(y-z) \cdot \frac{1}{x} + y(z-x) \cdot \frac{1}{y} + z(x-y) \cdot \frac{1}{z} = y-z + z-x + x-y = 0$$

Consider, $P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0$

$$\Rightarrow \log x + \log y + \log z = \log a \Rightarrow xyz = a \Rightarrow U = xyz$$

Choose $P_2 = 1$, $Q_2 = 1$, $R_2 = 1$

$$PP_2 + QQ_2 + RR_2 = x(y-z) + y(z-x) + z(x-y) = xy - xz + yz - yx + zx - zy = 0$$

Consider, $P_2 dx + Q_2 dy + R_2 dz = 0 \Rightarrow dx + dy + dz = 0$

$$\Rightarrow x + y + z = b \Rightarrow V = x + y + z$$

Solution is, $f(U, V) = 0$, i.e., $f(xyz, x+y+z) = 0$

6. A tightly stretched string of ~~length~~ with fixed end points $x=0$ & $x=l$ is initially in a position given by $u = v_0 \sin^3\left(\frac{\pi x}{l}\right)$, $0 \leq x \leq l$. If it is released from rest from this position, find the displacement function $u(x, t)$

Ans: Refer NB, Module - 2, page no. 10

7. Find the temperature of a rod of length 2m whose end points are maintained at temperature zero & the initial temperature is

$$P(x) = 100(2x - x^2)$$

Ans:

$$\text{Soln as, } u(x,t) = (C_1 \cos px + C_2 \sin px) e^{-c^2 p^2 t} \quad \text{--- (1)}$$

$$u(0,t) = u(2,t) = 0 \quad \text{--- (2)}$$

$$u(x,0) = 100(2x - x^2) \quad \text{--- (3)}$$

$$\text{From (2), } u(0,t) = 0$$

$$\Rightarrow C_1 = 0$$

$$\therefore \text{ soln becomes, } u(x,t) = C_2 \sin px e^{-c^2 p^2 t}$$

$$\text{From (2), } u(2,t) = 0$$

$$\text{Then } p = \frac{n\pi}{a}, \quad n \in \mathbb{Z}$$

$$\therefore \text{ soln becomes, } u(x,t) = C_2 \sin\left(\frac{n\pi x}{a}\right) e^{-c^2 \left(\frac{n\pi}{a}\right)^2 t}$$

$$\Rightarrow u(x,t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{a}\right) e^{-c^2 \left(\frac{n\pi}{a}\right)^2 t}$$

$$\text{From (3), } 100(2x - x^2) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{a}\right)$$

By Fourier Sine series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right), \quad \text{where } b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\text{Here } L = 2, \quad f(x) = 100(2x - x^2)$$

$$b_n = \frac{2}{2} \int_0^2 100(2x - x^2) \sin\left(\frac{n\pi x}{2}\right) dx = 100 \int_0^2 (2x - x^2) \sin\left(\frac{n\pi x}{2}\right) dx$$

$$u = 2x - x^2 \quad \quad \quad v = \sin\left(\frac{n\pi x}{2}\right)$$

$$u' = 2 - 2x \quad \quad \quad v_1 = -\cos\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right)$$

$$u'' = -2 \quad \quad \quad v_2 = -\sin\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right)^2$$

$$u''' = 0 \quad \quad \quad v_3 = \cos\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right)^3$$

$$b_n = 100 \left[\begin{aligned} & -(2x - x^2) \cos\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right) + (2 - 2x) \sin\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right)^2 \\ & - 2 \cos\left(\frac{n\pi x}{2}\right) \left(\frac{n\pi}{2}\right)^3 \end{aligned} \right]_0^2$$

$$b_n = 100 \left[(-0 + 0 - 2 \cos(n\pi) \left(\frac{n\pi}{2}\right)^3) - (0 + 0 - 2 \cos 0 \left(\frac{n\pi}{2}\right)^3) \right]$$

$$b_n = 100 \left[-2(-1)^n \left(\frac{n\pi}{2}\right)^3 + 2 \left(\frac{n\pi}{2}\right)^3 \right] = 200 \left(\frac{n\pi}{2}\right)^3 [1 - (-1)^n]$$

$$\therefore u(x,t) = \sum_{n=1}^{\infty} 200 \left(\frac{n\pi}{2}\right)^3 (1 - (-1)^n) \sin\left(\frac{n\pi x}{2}\right) e^{-c^2 \left(\frac{n\pi}{2}\right)^2 t}$$

DECEMBER 2019

PDE

1. Find the pde by eliminating the arbitrary function ϕ from $xyz = \phi(x+y+z)$.

Ans: Refer Qno: 1, Question paper, April 2018

2. Write any 3 assumptions involved in the derivation of 1-D wave equation

Ans: Refer NB, Module 2, page no: 2

3. Write all possible solutions of 1-D heat equation

Ans: Refer NB, Module 2, page no: 36 (Write the 3 solutions)

4. Find the general solution of $x^2(y-z)p + y^2(z-x)q = (x-y)z^2$

Ans: This is of the form $Pp + Qq = R$, where $P = x^2(y-z)$, $Q = y^2(z-x)$, $R = (x-y)z^2$

$$\text{A.C.} \Rightarrow \frac{dx}{P} = \frac{dy}{Q} = \frac{dz}{R} \Rightarrow \frac{dx}{x^2(y-z)} = \frac{dy}{y^2(z-x)} = \frac{dz}{z^2(x-y)}$$

$$\text{Let } P_1 = \frac{1}{x^2}, \quad Q_1 = \frac{1}{y^2}, \quad R_1 = \frac{1}{z^2}$$

$$\text{Then, } PP_1 + QQ_1 + RR_1 = x^2(y-z) \frac{1}{x^2} + y^2(z-x) \frac{1}{y^2} + z^2(x-y) \frac{1}{z^2} \\ = y-z + z-x + x-y = 0.$$

$$\text{Consider } P_1 dx + Q_1 dy + R_1 dz = 0 \Rightarrow \frac{1}{x^2} dx + \frac{1}{y^2} dy + \frac{1}{z^2} dz = 0$$

$$\text{Integrating, } -\frac{1}{x} - \frac{1}{y} - \frac{1}{z} = C_1 \Rightarrow \frac{1}{x} + \frac{1}{y} + \frac{1}{z} = -C_1 = a$$

$$\Rightarrow U = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

$$\text{Let } P_2 = \frac{1}{x}, \quad Q_2 = \frac{1}{y}, \quad R_2 = \frac{1}{z}$$

$$\text{Then, } PP_2 + QQ_2 + RR_2 = x^2(y-z) \frac{1}{x} + y^2(z-x) \frac{1}{y} + z^2(x-y) \frac{1}{z} \\ = x(y-z) + y(z-x) + z(x-y) \\ = xy - xz + yz - xy + xz - yz = 0$$

$$\text{Consider } P_2 dx + Q_2 dy + R_2 dz = 0 \Rightarrow \frac{1}{x} dx + \frac{1}{y} dy + \frac{1}{z} dz = 0.$$

$$\text{Integrating, } \log x + \log y + \log z = \log a \Rightarrow \log xyz = \log a \\ \Rightarrow v = xyz$$

$$\text{solution is, } f\left(\frac{1}{x} + \frac{1}{y} + \frac{1}{z}, xyz\right) = 0$$

5. Find the deflection of the vibration of a string which is fixed at the ends $x=0$ & $x=2$ & the motion is started by displacing the string into the form $\sin^2\left(\frac{\pi x}{2}\right)$ & released it with zero initial velocity at $t=0$

Ans: Refer NB, Module 2, page no: 10 (change $l=2$ & $y = \sin^2\left(\frac{\pi x}{2}\right)$)

6. Find the temperature distribution in a rod of length 2m whose end points are maintained at temperature zero & initial temperature is $f(x) = 100(2x - x^2)$

Ans: Refer Qno. 7, Dec 2018 Qppr

MODEL QUESTION PAPER

MAY 2019

PDE

1. Find the pde of $z = f\left(\frac{xy}{z}\right)$ by eliminating the arbitrary function

Ans: Refer NB, Module -1, page no: 24

2. Find the pde of all spheres having their centre on z -axis.

Ans: Refer NB, Module -1, page no: 31

3. Solve $\frac{\partial u}{\partial x} = 4 \frac{\partial u}{\partial y}$, $u(0,y) = 8e^{-3y}$ using method of separation of variables.

Ans: Let $U = U(x,y) = X Y$.

$$\frac{\partial U}{\partial x} = X' Y \quad \& \quad \frac{\partial U}{\partial y} = X Y'$$

$$\Rightarrow X' Y = 4 X Y' \Rightarrow \frac{X'}{X} = \frac{4 Y'}{Y} = k$$

Case 1: $\frac{X'}{X} = k \Rightarrow X' = kX$

put $X' = m$, $X=1 \Rightarrow m = k \Rightarrow X = C_1 e^{kx}$

Case 2: $\frac{4 Y'}{Y} = k \Rightarrow Y' = \frac{k Y}{4}$

put $Y' = m$, $Y=1 \Rightarrow m = \frac{k}{4}$

$\therefore U(x,y) = C_1 C_2 e^{kx} e^{\frac{k y}{4}}$

$U(0,y) = 8e^{-3y} \Rightarrow C_1 C_2 e^{\frac{k y}{4}} = 8e^{-3y} \Rightarrow C_1 C_2 = 8, \frac{k}{4} = -3 \Rightarrow k = -12$

$\therefore U(x,y) = 8 e^{-12x} e^{-3y} = 8e^{-(12x+3y)}$

4. Write down the 3 possible solutions of 1-D heat equation

Ans: Refer NB, Module 2, page no: 36 (Write the 3 solutions)

5. (a) Solve $\frac{y^2}{x} p + xzq = y^2$

(b) Find the pde of all planes which are at a constant distance k from the origin.

Ans: (a) Refer NB, Module -1, page no: 14

(b) Refer NB, Module -1, page no: 17

6. Solve: $x^2(y-z)p + y^2(z-x)q = z^2(x-y)$

Ans: Refer NB, Module -1, page no: 62

7. Find the temperature distribution in a rod of length a m whose end points are maintained at temperature 0°C & the initial temperature is $f(x) = 100(2x - x^2)$ $0 \leq x \leq a$.

Ans: Refer Appx, Dec 2019, Qno: 7

8. A string is stretched b/w 2 fixed points at a distance of 60cm & the points of the string are given initial velocities where $v = \begin{cases} \frac{\lambda x}{30}, & 0 < x < 30 \\ \frac{\lambda}{30}(60-x), & 30 < x < 60 \end{cases}$ x being the distance from an end. Find the displacement at any time t

Ans: Solution is, $y(x,t) = (c_1 \cos px + c_2 \sin px)(c_3 \cos cpt + c_4 \sin cpt)$ — (1)

Initial Boundary conditions: $y(0,t) = y(60,t) = 0$ — (2)

Initial conditions: $y(x,0) = 0$ — (3)
 $\frac{\partial y}{\partial t} \Big|_{t=0} = v = \begin{cases} \frac{\lambda x}{30}, & 0 < x < 30 \\ \frac{\lambda}{30}(60-x), & 30 < x < 60 \end{cases}$ — (4)

From (2), $y(0,t) = 0 \Rightarrow c_1 = 0$

solution becomes, $y(x,t) = c_2 \sin px (c_3 \cos cpt + c_4 \sin cpt)$

From (2), $y(60,t) = 0 \Rightarrow p = \frac{n\pi}{60}$

solution becomes, $y(x,t) = c_2 \sin\left(\frac{n\pi x}{60}\right) \left[c_3 \cos\left(\frac{n\pi ct}{60}\right) + c_4 \sin\left(\frac{n\pi ct}{60}\right) \right]$

$$\frac{\partial y}{\partial t} = c_2 \sin\left(\frac{n\pi x}{60}\right) \left[-c_3 \sin\left(\frac{n\pi ct}{60}\right) \cdot \frac{n\pi c}{60} + c_4 \cos\left(\frac{n\pi ct}{60}\right) \cdot \frac{n\pi c}{60} \right]$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = c_2 \sin\left(\frac{n\pi x}{60}\right) \left[c_4 \frac{n\pi c}{60} \right]$$

From (3), $y(x, 0) = 0$

$$c_2 \sin\left(\frac{n\pi x}{60}\right) [c_3 + 0] = 0 \Rightarrow c_2 c_3 \sin\left(\frac{n\pi x}{60}\right) = 0 \Rightarrow c_3 = 0$$

∴ solution becomes,

$$y(x, t) = c_2 \sin\left(\frac{n\pi x}{60}\right) c_4 \sin\left(\frac{n\pi c t}{60}\right) = c_2 c_4 \sin\left(\frac{n\pi x}{60}\right) \sin\left(\frac{n\pi c t}{60}\right)$$

$$y(x, t) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{60}\right) \sin\left(\frac{n\pi c t}{60}\right)$$

$$\frac{\partial y}{\partial t} = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{60}\right) \cos\left(\frac{n\pi c t}{60}\right) \cdot \frac{n\pi c}{60}$$

$$\left(\frac{\partial y}{\partial t}\right)_{t=0} = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{60}\right) \frac{n\pi c}{60} = \frac{\pi c}{60} \sum_{n=1}^{\infty} n b_n \sin\left(\frac{n\pi x}{60}\right)$$

$$\text{From (4), } \frac{\pi c}{60} \sum_{n=1}^{\infty} n b_n \sin\left(\frac{n\pi x}{60}\right) = v = \begin{cases} \frac{\lambda x}{30}, & 0 < x < 30 \\ \frac{\lambda}{30}(60-x), & 30 < x < 60 \end{cases}$$

$$\text{i.e., } \sum_{n=1}^{\infty} n b_n \sin\left(\frac{n\pi x}{60}\right) = \frac{60}{\pi c} \begin{cases} \frac{\lambda x}{30}, & 0 < x < 30 \\ \frac{\lambda}{30}(60-x), & 30 < x < 60 \end{cases}$$

By Fourier Sine series,

$$f(x) = \sum_{n=1}^{\infty} b_n \sin\left(\frac{n\pi x}{L}\right) \quad \text{where } b_n = \frac{2}{L} \int_0^L f(x) \sin\left(\frac{n\pi x}{L}\right) dx$$

$$\text{here } f(x) = \frac{60}{\pi c} \begin{cases} \frac{\lambda x}{30}, & 0 < x < 30 \\ \frac{\lambda}{30}(60-x), & 30 < x < 60 \end{cases}$$

$$L = 60$$

$$b_n \rightarrow n b_n$$

$$\therefore n b_n = \frac{2}{60} \left[\int_0^{30} \frac{60}{\pi c} \cdot \frac{\lambda x}{30} \cdot \sin\left(\frac{n\pi x}{60}\right) dx + \int_{30}^{60} \frac{60}{\pi c} \cdot \frac{\lambda}{30} (60-x) \sin\left(\frac{n\pi x}{60}\right) dx \right]$$

$$= \frac{2}{60} \times \frac{\lambda}{30} \left[\int_0^{30} x \sin\left(\frac{n\pi x}{60}\right) dx + \int_{30}^{60} (60-x) \sin\left(\frac{n\pi x}{60}\right) dx \right]$$

$$= \frac{\lambda}{15\pi c} \left[\left(-x \cos\left(\frac{n\pi x}{60}\right) \cdot \frac{60}{n\pi} + \sin\left(\frac{n\pi x}{60}\right) \left(\frac{60}{n\pi}\right)^2 \right) \Big|_0^{30} + \left(-(60-x) \cos\left(\frac{n\pi x}{60}\right) \cdot \frac{60}{n\pi} - \sin\left(\frac{n\pi x}{60}\right) \left(\frac{60}{n\pi}\right)^2 \right) \Big|_{30}^{60} \right]$$

$$u = x \quad v = \sin\left(\frac{n\pi x}{60}\right)$$

$$u' = 1 \quad v_1 = -\cos\left(\frac{n\pi x}{60}\right) \cdot \frac{60}{n\pi}$$

$$u'' = 0 \quad v_2 = -\sin\left(\frac{n\pi x}{60}\right) \left(\frac{60}{n\pi}\right)^2$$

$$u = 60-x \quad v_1 = -\cos\left(\frac{n\pi x}{60}\right) \cdot \frac{60}{n\pi}$$

$$u' = -1 \quad v_2 = -\sin\left(\frac{n\pi x}{60}\right) \left(\frac{60}{n\pi}\right)^2$$

$$u'' = 0$$

$$n b_n = \frac{\lambda}{15\pi c} \left[\left(-30 \cos\left(\frac{n\pi}{2}\right) \frac{60}{n\pi} + \sin\left(\frac{n\pi}{2}\right) \left(\frac{60}{n\pi}\right)^2 - 0 \right) + \left(0 - (-30 \cos\left(\frac{n\pi}{2}\right) \frac{60}{n\pi} - \sin\left(\frac{n\pi}{2}\right) \left(\frac{60}{n\pi}\right)^2 \right) \right]$$

$$nb_n = \frac{\lambda}{15\pi c} \left[-30 \cos\left(\frac{n\pi}{2}\right) \frac{60}{n\pi} + \sin\left(\frac{n\pi}{2}\right) \left(\frac{60}{n\pi}\right)^2 + 30 \cos\left(\frac{n\pi}{2}\right) \frac{60}{n\pi} + \sin\left(\frac{n\pi}{2}\right) \left(\frac{60}{n\pi}\right)^2 \right]$$

$$nb_n = \frac{\lambda}{15\pi c} \cdot 2 \sin\left(\frac{n\pi}{2}\right) \left(\frac{60}{n\pi}\right)^2 \Rightarrow b_n = \frac{2\lambda}{15\pi^3 n^3 c} \sin\left(\frac{n\pi}{2}\right) 60^2 \times 60$$

$$\Rightarrow b_n = \frac{2\lambda}{15\pi^3 n^3 c} \sin\left(\frac{n\pi}{2}\right) 2160 = \frac{480\lambda}{\pi^3 n^3 c} \sin\left(\frac{n\pi}{2}\right)$$

$$\therefore y(x,t) = \sum_{n=1}^{\infty} \frac{480\lambda}{\pi^3 n^3 c} \sin\left(\frac{n\pi}{2}\right) \sin\left(\frac{n\pi x}{60}\right) \sin\left(\frac{n\pi c t}{60}\right)$$

MODEL QUESTION PAPER

PDE

MAY/JUNE 2016

1. Obtain the pde by eliminating f & g from $z = x f(y) + y g(z)$

Ans: Refer NB, Module -1, page no: 20

2. Solve the pde $(y^2 + z^2)p - xyq + xz = 0$.

Ans: Refer Qno: 3, Qpp: July 2018

3. Obtain the solution of the wave equation $\frac{\partial^2 u}{\partial x^2} = \frac{1}{c^2} \frac{\partial^2 u}{\partial t^2}$ using method of

separation of variables where the separation constant $k < 0$.

Ans: Refer NB, Module -2, page no: 4 (Case: k negative for both x & t)

4. Write any 2 assumptions involved in deriving 1-D wave equation

Ans: Refer NB, Module -2, page no: 2

5. Solve $\frac{\partial u}{\partial t} = h \frac{\partial^2 u}{\partial x^2}$ subject to conditions $u(0,t) = u(1,t) = 0$ for $t > 0$ &

$$u(x,0) = 3 \sin n\pi x, \quad 0 < x < 1$$

Ans: Solution is, $u(x,t) = (C_1 \cos px + C_2 \sin px) e^{-h^2 p^2 t}$

$$\text{Boundary conditions: } u(0,t) = u(1,t) = 0 \quad \text{--- (1)}$$

$$\text{Initial conditions: } u(x,0) = 3 \sin n\pi x \quad \text{--- (2)}$$

$$\text{(1)} \Rightarrow u(0,t) = 0$$

$$\Rightarrow C_1 = 0$$

$$\text{Solution becomes, } u(x,t) = C_2 \sin px e^{-h^2 p^2 t}$$

From ①, $u(1,t) = 0 \Rightarrow p = n\pi, n \in \mathbb{Z}$

Solution becomes, $u(x,t) = c_2 \sin(n\pi x) e^{-h^2 n^2 \pi^2 t}$

From ②, $u(x,t) = \sum_{n=1}^{\infty} b_n \sin(n\pi x) e^{-h^2 n^2 \pi^2 t}$

From ③, $u(x,0) = 3 \sin n\pi x$

$$\Rightarrow \sum_{n=1}^{\infty} b_n \sin n\pi x = 3 \sin(n\pi x)$$

$$\Rightarrow b_n = 3$$

$$\therefore u(x,t) = 3 \sum_{n=1}^{\infty} \sin(n\pi x) e^{-h^2 n^2 \pi^2 t}$$

6. Form the pde by eliminating a, b, c from $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Ans: Refer NB, Module 1, page no: 14.

7. Solve $x(y^2 - z^2)p + y(z^2 - x^2)q = z(x^2 - y^2)$

Ans: Refer NB, Module-1, page no: 53

8. A tightly stretched of length 'a' with fixed ends is initially in equilibrium position. Find the displacement $u(x,t)$ of the string if its set vibrating by giving each of its points a velocity $V_0 \sin\left(\frac{n\pi x}{a}\right)$

Ans: Refer NB, Module-2, page no: 15 (Take velocity $V_0 \sin\left(\frac{n\pi x}{a}\right)$)

9. A transversely vibrating string of length 'a' is stretched b/w 2 points A & B. The initial displacement of each point of the string is zero & the initial velocity at a distance x from A is $kx(a-x)$. Find the form of the string at any subsequent time.

Ans: Refer NB, Module-2, page no: 12

10. Find the temperature in a laterally insulated bar of length L whose ends are kept at temperature zero if the initial temperature is

$$f(x) = \begin{cases} x, & 0 < x < L/2 \\ L-x, & L/2 < x < L \end{cases}$$

Ans: Refer Qppr, July 2017, Qno: 10.